

COMP3011J MC Project (Beta)

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1 Brief Overview of the Open-Source App: **Kandean**

With the original version of **Kandean**, users can effortlessly create, post, and share their unique recipes, meanwhile, comment, likes, and save various recipes.

The coming version of this app will also provide a digital fridge that tracks available ingredients. The possible coming features are: timely reminders for items nearing expiration, setting automatic updates for the quantity of each item stored and receive shopping suggestions when ingredients run low.

While most of the popular recipe applications focus on creating and sharing recipes or diet planning, **Kandean** uniquely integrates these functionalities with a brand-new digital fridge feature. Having this, you can easily manage your diet and shopping choices.

2 Newly Implemented Feature: Auto-decreasing Ingredients

Add a function that automatically decreases the quantity of an ingredient in the digital fridge when it is used in a recipe. This is achieved by clicking "USED" button of the ingredient used and it will update the quantity of the item or delete it if used up.

The corresponding changes are made in:

2.1 Recipe Detail View 1

For the display of the ingredients in *fragment_digital_recipe.xml*, I replace the original simple text-view with a well-formatted recycler-view to show ingredients in a list, which containing a "USED" button besides each (in *item_ingredient.xml*). By clicking the button, the corresponding ingredient's current quantity will decrease accordingly.

To handle potential unit incompatibilities (e.g., "g" vs. "kg"), we compare and unify units using the *getConversionFactor* function in the *IngredientAdapter* class. There are various cases for auto-decreasing:

- If the current storage is sufficient, only the item's quantity is updated.
- If the quantity equals the current stock, the item is deleted.
- If there is insufficient stock, the system uses the available quantity first, then searches for more of the ingredient and performs the update again.

This is managed by the *processItemUpdate* function in the *IngredientAdapter* class.

2.2 Create Recipe Functionality 2

To ensure that recipe providers specify the quantity and unit of each ingredient, I have adjusted the create recipe page. Each ingredient's information must be entered in a row containing the name, quantity, and unit. Units can only be selected from provided options. And if you want to add more, you need to click the add button. A text hint is provided for guidance below this section. This part is achieved in the *CreateRecipeFragment* class and *fragment_create_recipe.xml*.

Additionally, the outdated image selection, upload, and encoding algorithm has been replaced with a current algorithm that can be used.

2.3 Edit Recipe Functionality

Nearly the same with create part.

3 Newly Implemented Feature: Storage Category Filter 3

In the Digital Fridge Display page, I add a category tab under the search bar, allowing users to click different category tabs to display corresponding items.

4 UI Page Refinements

I have redesigned the bottom page-switching bar based on user preferences and habits, and fine-tuned parameters on other pages for consistent style and size.

5 Development Timeline 1

Time	Work to be completed	Comments
ALPHA SUBMISSION	Set up the new basic pages and layout of the digital fridge page on the main page	Basic new function setup based on the existing app
BETA SUBMISSION	Add the function of automatically decreasing the quantity of an item in the digital fridge if used in the recipe	Link the digital fridge with the recipe usage
FINAL SUBMISSION	Add an API that enables users to scan barcodes or items to add new items more easily	Add APIs

Table 1: Timeline for implementation

Figure 1: Left: Original Detail Recipe Page; Right: Current

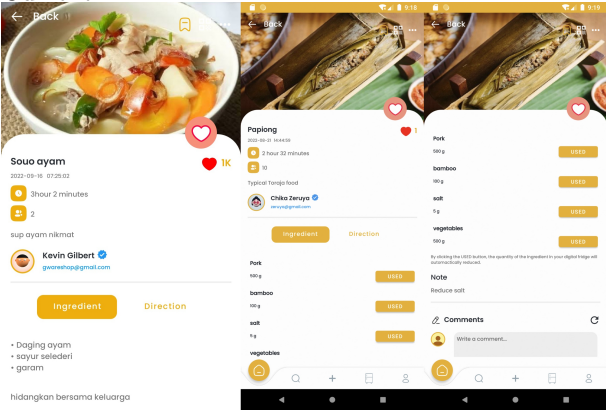


Figure 2: Left: Original Create Recipe Page; Right: Current

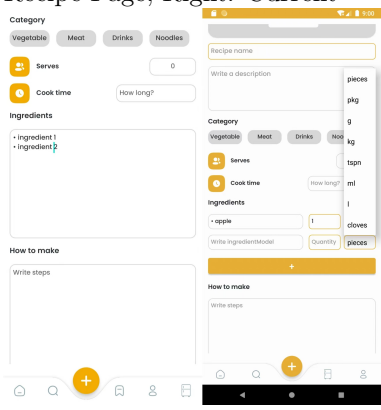


Figure 3: Left: Original Digital Fridge Page; Right: Current

